

RA-02: Lean verification of the negative polynomial-factor result

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Scope and attribution. Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge, retains mathematical authorship of the original negative resolution. His manuscript *Sharp worst-case factors for randomly pivoted Cholesky and LU*, Theorem 1 and Corollary 3, remains unchanged. This note records the Lean formalization and its verification. Chen, Epperly, Tropp and Webber retain credit for RPCholesky and the comparison problem. Substantial OpenAI Codex assistance is disclosed.

Complete original target

For a complex Hermitian positive-semidefinite matrix $A \in \mathbb{C}^{n \times n}$, set $R_0 = A$. At a nonzero residual, select j with conditional probability $(R_t)_{jj} / \text{tr}(R_t)$ and set

$$R_{t+1} = R_t - \frac{R_t(:, j)R_t(j, :)}{(R_t)_{jj}}.$$

Zero residuals remain zero; a zero diagonal has zero probability at a nonzero residual. Let $\lambda_1(A) \geq \dots \geq \lambda_n(A) \geq 0$ be the actual ordered eigenvalues and $\tau_r(A) = \sum_{j>r} \lambda_j(A)$.

The theorem `NLA.RA02.no_polynomial_trace_factor` proves that there are no real constants $C > 0$ and $p \geq 0$, independent of n, r, A , such that

$$\mathbb{E} \text{tr}(R_r) \leq Cr^p \tau_r(A)$$

for every $n \geq 1$, every complex Hermitian positive-semidefinite A , and every integer $1 \leq r \leq n$, after exactly r pivots. The stronger constructive export `NLA.RA02.universal_counterexamples` supplies, for every such real C, p , a rank parameter $r \geq 1$ and an actual positive-definite witness $A \in \mathbb{C}^{(r+1) \times (r+1)}$ with

$$0 < \tau_r(A), \quad Cr^p \tau_r(A) < \mathbb{E} \text{tr}(R_r).$$

The formal process includes every ordered history, including repeated and zero-probability choices. Its finite path weights are proved nonnegative and normalized, with the actual conditional recursion. Uniform dummy labels after absorption do not change the zero residual. The tail uses Mathlib's genuinely decreasing Hermitian eigenvalues, with a proved Euclidean Rayleigh bound; it is not defined by the counterexample's desired estimate.

Finite proof and computation

For $r \geq 1$, the proof uses an explicit real-entry arrowhead matrix of order $r + 1$. With

$$t = \frac{1}{2r+1}, \quad \varepsilon = t^{2(r+1)}, \quad d_i = \varepsilon^i, \quad \alpha_i = t^{i+1} \quad (0 \leq i < r),$$

and $D = \text{diag}(d_0, \dots, d_{r-1})$, it is

$$A = \begin{pmatrix} D & D\alpha \\ \alpha^T D & \varepsilon^r + \alpha^T D\alpha \end{pmatrix}.$$

Its exact complex quadratic form proves positive definiteness. A nonzero Rayleigh probe gives $0 < \tau_r(A) \leq \varepsilon^r$. Exact residual-state identities, positive prefix traces and an injective binary history encoding give

$$\mathbb{E} \text{tr}(R_r) \geq \frac{2^r \varepsilon^r}{(1 + 1/r)^r} \geq \frac{2^r}{3} \tau_r(A).$$

The retained 2^r histories lower-bound the full normalized expectation; they are not renormalized into a different sampling law. Exponential domination of every real power then yields the strict violations above.

The only numerical certificate is $\exp(1) \leq 3$, proved by LeanCert with explicit kernel trust. It is consumed by `rank_denominator_le_three`, then `exponential_tail_factor` and the final counterexample chain. All rank, history, matrix and real-exponent calculations remain symbolic; there is no interval subdivision or numerical eigenvalue approximation.

The sufficient factor $2^r/3$ resolves the complete original question. The manuscript's sharper limiting ratio 2^r , entrywise-positive and correlation-matrix extensions, its separate LU results, and oversampling are not additional formal claims of this project.

Exact verification evidence

The immutable proof revision `26dc080e47b7` passed actual non-root Linux run `35175272827`, job `105055587640`. All 220 submitted inputs were authenticated. The default kernel and Comparator accepted all 27 frozen contracts, with only `propext`, `Classical.choice` and `Quot.sound`. The required rejection, regression and build/export sandbox controls passed.

The project pins Lean 4.33.1, Mathlib revision `0df444a360ea` and LeanCert revision `621a43d7cf21`. Complete hashes, commands, logs, source correspondence and review scopes are retained with the canonical Lean project. With pinned dependencies, `lake build` is its local entry point; its README gives the complete Linux verification commands.

Two complete nonauthor proof-source reviews and a separate independent runtime evidence audit are retained. They are AI-agent reviews, not external human peer review or official Tau Ceti endorsement. The earlier serial macOS development and the GitHub Linux run are distinct executions; evidence audits are not further compiler runs. Publication-commit and upstream PR checkouts require their own applicable checks, which are not claimed by this note.